Assignment 1: DEEP LEARNING

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Question 1

Consider the function:

$$f(w) = (w-4)^2$$

Task 1 (Derivative)

- Compute the derivative $f'(w)$.

Task 2 (Gradient Descent Updates)
Starting from $w_0 = 1$ and learning rate $\eta = 0.2$:

$$w_{t+1} = w_t - \eta f'(w_t)$$

- Compute the first **ten** gradient descent updates.
- Show each computation step clearly (derivative $\rightarrow$ update $\rightarrow$ new value).

Question 2

Write a small program (Python) that:

- Defines the function
$$f(w) = (w-3)^2.$$
- Computes the derivative
$$f'(w) = 2(w-3).$$
- Applies **manual gradient descent** for 10 iterations with learning rate
$$\eta = 0.1.$$
- Prints the updated value of w at each iteration.

Questions:

- What value does w approach and why?
- Why is this a good toy example for understanding gradient descent?

Question 3

Extend your gradient descent program to:

1. Plot the trajectory of w_t over the 10 iterations.
2. Plot the value of the function $f(w_t)$ vs. iteration number.
3. Try different learning rates $\eta = 0.01, 0.1, 0.5, 1.0$.

Questions to Answer:

- How does the learning rate affect speed of convergence?
- When does the algorithm diverge?
- What does this teach you about choosing a good learning rate in deep learning?

Question 4

Consider the function:

$$g(w) = w^4 - 3w^3 + 2$$

Task:

- Compute the derivative $g'(w)$.
- Apply gradient descent for 15 iterations starting at $w_0 = 0.5$ with $\eta = 0.05$.
- Plot your results.

Reflection:

- Why do non-convex functions make optimization harder?
- How does this relate to deep learning loss landscapes?

Question 5

Consider a deep network with two hidden layers.

- Input:

$$x = [1.2,\ -0.7,\ 0.5,\ 2.0]$$

- Layer 1 (3 neurons):

$$W_1 = \begin{bmatrix} 0.2 & -0.1 & 0.4 & 0.3 \\ -0.5 & 0.2 & 0.1 & -0.3 \\ 0.7 & -0.4 & 0.6 & 0.1 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 0.05 \\ -0.02 \\ 0.10 \end{bmatrix}$$

- Layer 2 (2 neurons):

$$W_2 = \begin{bmatrix} 0.3 & -0.5 & 0.2 \\ -0.6 & 0.1 & 0.4 \end{bmatrix}, \quad b_2 = \begin{bmatrix} 0.01 \\ -0.03 \end{bmatrix}$$

- Output layer:

$$W_3 = [0.4 \quad -0.2], \qquad b_3 = 0.1$$

- Activation: Sigmoid

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

- Target:

$$y = 1 \qquad L = \frac{1}{2}(\hat{y} - y)^2$$

- Learning rate:

$$\alpha = 0.05$$

Compute the Gradient Descent for Parameter Updates

$$W_i^{new} = W_i - \alpha \frac{\partial L}{\partial W_i}, \qquad b_i^{new} = b_i - \alpha \frac{\partial L}{\partial b_i}$$

Task:

- Perform Forward Pass and Backpropagation
- Explain the meaning of $\frac{\partial L}{\partial W_3}$.
- How does Layer 2 influence gradient flow?
- What happens if Layer 1 activations are near zero?
- Explain the effect of a large learning rate α?